

HS 123 Quiz 1

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1

Given 4 options a, b, c, d person 1's payoff function $u(\cdot)$ gives values $u(a) = 100$, $u(b) = 0.003$, $u(c) = 0.001$, $u(d) = 0$. Person 2's payoff function $v(\cdot)$ gives values $v(a) = 0.112$, $v(b) = 0.111$, $v(c) = 0.110$, $v(d) = -0.13$. The preference is ordinal. (1 Point)

- ☐ 1 and 2 have different preferences because for 1 a is much more preferred over b, c, and d.
- ☐ 1 and 2 have different preferences because 2 prefers a, b, c, d much less than 1 does.
- ☐ 1 and 2 have different preferences because 2 has a negative payoff.
- ☐ 1 and 2 have same preference over a, b, c, d.

2

Some money can be given to 1 and 2. Any allocation can be represented by a pair (x, y) , where x and y are the money given to 1 and 2 respectively. Suppose 1's preference is such that he likes to have more money but he does not like if 2 gets more money. 1's payoff function can be represented by, (1 Point)

- ☐ $u(x, y) = x+y$
- ☐ $u(x, y) = x-y$
- ☐ $u(x, y) = xy$
- ☐ $u(x, y) = 10x$

3

Some money can be given to 1 and 2. Payoff function of 1 is $u(x, y) = xy$, where x and y are the money given to 1 and 2 respectively. Suppose a, b, c are the three allocations with $a = (10, 10)$, $b = (4, 16)$, $c = (15, 5)$. How does 1 rank a, b and c ? (1 Point)

- ☐ c, a, b
- ☐ a, c, b
- ☐ b, c, a
- ☐ a, b, c

4

In a coastal village there are n fishermen. Each has the option to fish more (M) or fish less (L). If all choose M, fish in the sea cannot sustainably breed and soon there will be no fish. If all choose L then the fish can breed. So, for you (a fisherman) an outcome where everyone chooses L is preferred to everyone choosing M. If all but you choose L, you would like to choose M (rather than L) because that gives you more fish. If all else choose M, you would like to choose M because choosing L means less fish. This situation resembles, (2 Points)

- ☐ Pure coordination game
- ☐ Matching pennies game
- ☐ Battle of sexes game
- ☐ Prisoner's dilemma game

5

The Nash equilibrium/equilibria of this game is/are, (2 Points)

		player 2		
		L	M	R
player 1	U	5,3	7,2	2,1
	C	1,2	6,3	1,4
	D	4,5	6,4	3,5

- ☐ (U, L) only
- ☐ There is no Nash equilibrium in this game
- ☐ (U, L), (D, R)
- ☐ (U, M) only
- ☐ (C, M) only

6

In the following game the Nash equilibrium/equilibria is/are, (2 Points)

		Player b		
		L	C	R
Player a	U	4,1	4,6	3,2
	M	2,0	5,1	1,2
	D	3,3	4,2	4,1

- ☐ (U,C) only
- ☐ (U,C), (D,L)
- ☐ (D,L), (D,C)

☐ There is no Nash equilibrium

7

Three friends A, B and C have to decide whether to contribute or not to contribute a fixed amount of money for building a park. Their preferences for different action profiles are represented by the payoff matrices given below. A chooses the row, B chooses the column, C chooses the matrix. Their payoffs are shown in A,B,C order. The Nash equilibrium of the game is, (2 Points)

		B				B	
		Contribute	Don't Contribute			Contribute	Don't Contribute
A	Contribute	5,5,5	3,6,3	A	Contribute	3,3,6	1,4,4
	Don't Contribute	6,3,3	4,4,1		Don't Contribute	4,1,4	2,2,2
		Contribute	← C →			Don't Contribute	

- ☐ (Contribute, Contribute, Don't Contribute),
- ☐ (Don't Contribute, Contribute, Don't Contribute)
- ☐ (Contribute, Contribute, Contribute)
- ☐ (Don't Contribute, Don't Contribute, Don't Contribute)

8

The game in question 7 resembles (1 Point)

- ☐ Battle of sexes game
- ☐ Prisoner's dilemma game
- ☐ Assurance game
- ☐ Stag hunt game

9

Take the BoS game with a variation. H's preferences are like before. But W wants to avoid H. Meeting H is the worst outcome for her. If they don't meet she would rather be in Boxing Match rather than at Opera. The Nash equilibrium/equilibria of the game is/are (2 Points)

- ☐ (B, B), (O, O)
- ☐ (B, B) only
- ☐ (B, O) only
- ☐ There is no Nash equilibrium

10

The game in question 9 is like which game we discussed? (1 Point)

- ☐ Matching pennies

- ☐ Prisoner's dilemma

☐ FISHER'S GREETING

☐ "Which Tyre?" coordination game

☐ Stag hunt

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HS 123: Game Theory and Economics
B Tech, 4th Semester, Mid Semester Test
Total Marks: 40
Date: 6th March, 2022

[Explain the answers in words. Without proper elaboration of answers, marks would be deducted.]

1. In the following zero-sum game only the payoffs of player A are displayed. Using iterated elimination of strictly dominated actions try to solve it. Is/are the solution/s a Nash equilibrium/equilibria? [4+2]

		B		
		L	M	R
A	U	5	1	3
	S	6	1	2
	D	1	0	0

2. Suppose there are two aircraft manufacturers A and B , together they produce all aircrafts in the world. The price of new aircrafts is determined by the function, $P = 180 - Q$, where $Q = q_A + q_B$. Cost of building an aircraft is \$30 million for both companies. Or, $c_A = c_B = 30$, where the cost per aircraft is measured in millions of dollars.
- (a) Write the profit function of each company in terms of q_A and q_B as well as c_A and c_B . Find each company's best response function. [2+2 = 4]
- (b) From (a) above find the Nash equilibrium quantity of aircrafts a year produced by the two companies. What is the price of an aircraft? How much profit is made by each company. [2+2+2 = 6]
- (c) Suppose A has developed an improved technology to manufacture aircrafts due to which its cost is lower. The cost of B has however gone up. The cost per aircraft of two companies are now, $c_A = 20$, $c_B = 40$. What is the market share of each company? What are the profits of each company? [2+2 = 4]
3. 50 students in a class play the following game. Each chooses an integer from 0 to 100. Let X = the mean of the 50 chosen numbers. The student who has chosen a number which is closest to the two-thirds of X wins a prize of 100 rupees. If multiple students tie, they split the prize equally.
- (a) Show that choosing the number 80 is a weakly dominated action. [2]
- (b) Suppose all else have chosen 40, then for an individual player i , what is the set of best responses? [6]
- (c) Find a symmetric Nash equilibrium of the game (that is, a Nash equilibrium where everyone chooses the same number). [4]
4. In contrast to the Cournot model, suppose **prices** are decided by firms. Two stores are there, A (which sells bread) and B (which sells butter). The cost of producing a loaf of bread is 1 rupee and the cost of producing a packet (200 g) of butter is 2 rupees. If the price of A 's bread is P_1 per loaf, and price of B 's butter is P_2 per packet, then their

respective monthly sales (Q_1 thousand loaves of bread and Q_2 thousand packets of butter) are given by,

$$Q_1 = 14 - P_1 - 0.5 P_2 \text{ and}$$

$$Q_2 = 19 - 0.5 P_1 - P_2$$

- (a) For each store, write the profit as a function of P_1 and P_2 . [2]
- (b) Using (a), derive their best response functions. [3]
- (c) Using (b), derive Nash equilibrium prices. [3]

HS 123: Game Theory and Economics
BTech 4th Semester, Quiz 2
Total Marks 10, Time 55 minutes
Date: 9th April, 2022
[Answers should be accompanied by proper elaboration]

1. In a Bertrand duopoly market, firm 1 maximises its market share subject to not making a loss. Firm 2 maximises its profit. Their unit costs are same, c . The market demand function is given by, $D(p) = \alpha - p$, if $\alpha \geq p$; $D(p) = 0$, if $\alpha < p$; $\alpha > c$. Other standard assumptions as discussed in the class hold here. Find the best response functions of the firms and the Nash equilibrium/equilibria. [3+3]

2. Find *all* the Nash equilibria of a second-price sealed-bid auction with two bidders with the same setting as mentioned in the class. [4]

(Hint: construct best response functions.)

HS 123: Game Theory and Economics
BTech 4th Semester, End Semester Test
Total Marks 40, Date: 10th May, 2022

[Answers should be accompanied by proper elaboration]

1. Consider a Bertrand duopoly market as discussed in the class except that unit costs of firms are different: $c_1 = 2 < c_2 = 4$. The market demand function is, $D(p) = 10 - p$, if $10 \geq p$; $D(p) = 0$, if $10 < p$.
 - (a) Suppose that the rule of splitting up consumers when $p_1 = p_2 = 4$ assigns all consumers to firm 1; when $p_1 = p_2 \neq 4$ each firm gets half of consumers. Show that $(p_1, p_2) = (4, 4)$ is the unique Nash equilibrium of this game. What are the equilibrium profits? [4+2]
 - (b) Suppose the rule of splitting up consumers when $p_1 = p_2$ assigns half of consumers to each firm. Show that no Nash equilibrium exists in this case. [3]
2. [Electoral competition with discrete positions] Consider the Hotelling model as discussed in the class with the modification that there are only 101 voters with their favourite positions at -50, -49, ..., 49, 50. Two candidates (players) are competing.
 - (a) Write down the best response functions of the players. [3]
 - (b) What is/are the Nash equilibrium/equilibria? [2]
3. Consider the following game with von Neumann Morgenstern preferences.
 - (a) Is there any strict domination in pure strategies? [1]
 - (b) Is there any strict domination in mixed strategies? If yes, identify the dominated actions, and after eliminating them find mixed strategy Nash equilibria of the game. [3+3]

		Player 2		
		X	Y	Z
Player 1	A	1, 3	2, 3	1, 4
	B	2, 2.5	4, 0	0, 6
	C	1, 1.5	0, 4	4, 0

4. In a strategic game with von Neumann Morgenstern preferences where each player has the same set of actions, a *symmetric mixed strategy Nash equilibrium* is defined as one which is a mixed strategy Nash equilibrium and the mixed strategy is same for every player. In the following game, what is/are the symmetric mixed strategy Nash equilibrium/equilibria? [4]

		Player 2	
		A	B
Player 1	A	0, 0	1, 1
	B	1, 1	0, 0

5. [Stackelberg duopoly] The market demand function is $D(p) = 10 - p$, if $10 \geq p$; $D(p) = 0$, if $10 < p$. The unit cost of production of the two firms is 2. Now, consider a variation of the Cournot duopoly model. Both firms decide how much to produce (q_i), but they do it sequentially. First, firm 1 (the leader) decides q_1 , after which firm 2 (the follower) decides q_2 . Each firm likes to maximise its profit.

- (a) Specify this extensive game with perfect information. [2]
(b) Apply backward induction to find the subgame perfect equilibrium/equilibria of the game. [4]
(c) What are the profits of the firms in the subgame perfect equilibrium/equilibria? [4]
6. Consider the Ultimatum Game as discussed in the class. Player 1 offers $100 - \alpha$ to player 2 and keeps α to herself. Find the values of α for which there is a *Nash equilibrium*. [Note, while describing a Nash equilibrium, the strategies of players need to be specified] [5]

